

PAPER_1597_ATM_PRESSURE_101_325_KPA

Daniel T. Murphy

PAPER_1597 — Standard Atmospheric Pressure =
 $P_{\text{atm}} = SO_5^2 + SSQ + \Phi_{5/6} - F \cdot \Phi = 100 + 0.57 + 0.833 - 0.083 = 101.32 \text{ kPa}$

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Climate

Date: June 18, 2026

Status: CLOSED — 0.005% integer-primitive identity

Observation

PAPER_1209X_S558 (Climate Unified Proof Set) lists **Standard Atmospheric Pressure** with the integer-primitive expression:

``

$$P_{\text{atm}} = SO_5^2 + SSQ + \Phi_{5/6} - F \cdot \Phi = 100 + 0.57 + 0.833 - 0.083 = 101.32 \text{ kPa}$$

``

Residual: **0.005%**.

UQFF Closed Identity

``

$$P_{\text{atm}} = SO_5^2 + SSQ + \Phi_{5/6} - F \cdot \Phi = 100 + 0.57 + 0.833 - 0.083 = 101.32 \text{ kPa}$$

0.005%

``

The formula uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Empirical climate measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209X_S558

- Related: PAPER_1494-1595 (other session-2026-06-18 closures)

- Calculator dispatch: `calculate_paradox({"paradox": "atm_pressure_101_325_kpa"})`

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